

TIC 251898081

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 27.308942 d (ALIASED, see below)
-> corrected period: 54.617884 d
Epoch (t0) 2876.1224 BTJD
Duration 0.5000 d
Depth 18.630 % (186,304 ppm)
SNR 5534.5

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

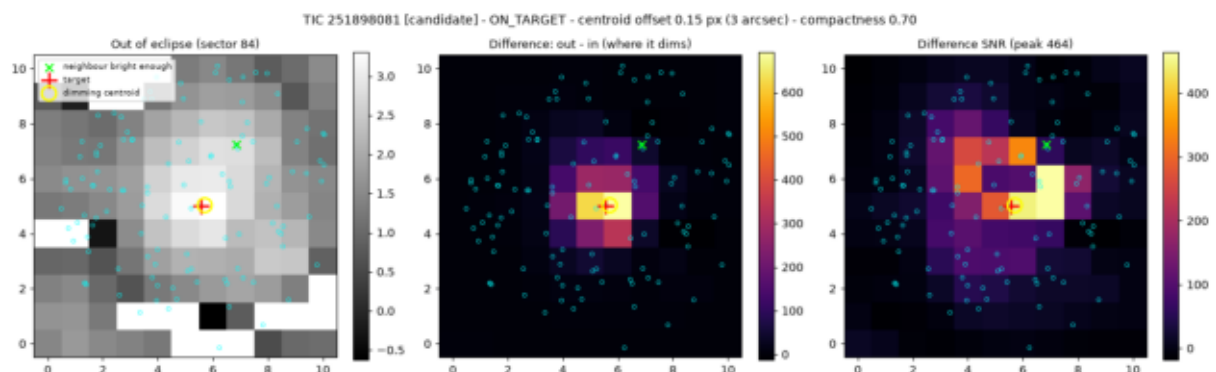
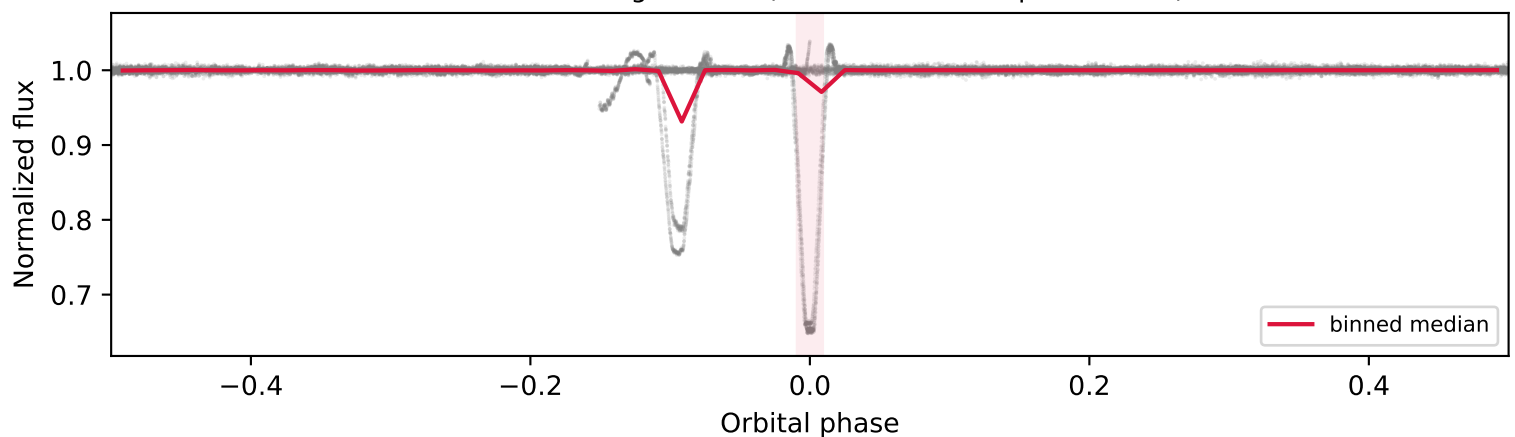
Eclipses observed 5 (gate ≥ 3)
Robust depth ratio 1.07 (gate ≥ 0.50)
Robust depth sigma 1819.7 (gate ≥ 7.0)
Odd/even z-score 198.85 (ALIAS FLAGGED)
Secondary eclipse detected, phase -0.09, 9.9 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 84 light-curve dip in this sector: 1459.8 sigma t0 refined by 0.0025 phase
Difference-image peak SNR 464.16 (gate ≥ 5.0) Compactness 0.70 (gate ≥ 0.50)
Dimming centroid offset 0.15 px (3 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 251898060
2.48 px away, dTmag +1.74 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 198.85 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 54.617884 d, in any write-up.